

1(a)	bronchiole ; I respiratory / terminal, before bronchiole	1
1(b)	(actual diameter) = image / observed, length, ÷ magnification ; A (A =) I ÷ M or magnification triangle 300 µm ; A 275 µm A 288 µm A 313 µm A 325 µm (12 000 / 40) (11 000 / 40) (11 500 / 40) (12 500 / 40) (13 000 / 40)	2

Question	Answer	Marks
1(c)	<i>any four from</i> <i>in healthy lungs</i> 1 correct direction of movement of <u>both</u> respiratory gases ; e.g. oxygen from alveolus towards blood and carbon dioxide from blood to alveolus ; oxygen enters the blood system and carbon dioxide leaves A red blood cell / haemoglobin, <i>as ref. to blood</i> 2 diffusion (of, oxygen / carbon dioxide) or movement, down a concentration gradient / from high(er) to low(er) concentration ; A implied e.g. oxygen enters blood from a <u>higher</u> concentration I diffusion of gases 3 detail of pathway ; R <i>ref. to</i> cell walls e.g. across, alveolar wall / squamous (epithelial) cells across endothelium / capillary wall ; A squamous cells <i>in context of capillary</i> crosses two layers of cells (<i>alveolar wall and capillary wall</i>) <i>comparison healthy with diseased – look for ora</i> 4 higher rate of exchange / increased rate of diffusion / steeper concentration gradient ; A more oxygen to blood per unit time / more carbon dioxide to alveolus per unit time I more efficient gas exchange I better gas exchange / faster diffusion 5, 6 AVP ; ; e.g. larger surface area (for, gas exchange / diffusion) shorter diffusion distance <i>ref. to</i> (greater) ability to, stretch / recoil (for ventilation to maintain gradient) or <i>ref. to</i> elasticity (<i>more v fewer elastic fibres is not sufficient</i>)	4